

MODULE : III Fourier Transform

Even Function: If $f(x)$ is said to be even function, then $f(-x) = f(x)$.

Examples: $x^2, \cos x, x \sin x$, etc.,

Geometrical Meaning

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

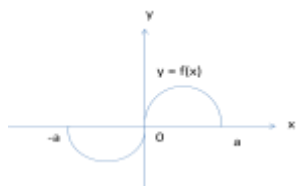


Odd Function: If $f(x)$ is said to be odd function, then $f(-x) = -f(x)$.

Examples: $x, \sin x, x^2 \sin x$, etc.,

Geometrical Meaning

$$\int_{-a}^a f(x) dx = 0$$



Modulus Function

If $f(x) = |x|$, $-\infty < x < \infty$, then $f(x)$ can be mathematically written as

$$f(x) = \begin{cases} -x, & -\infty < x < 0 \\ x, & 0 < x < \infty \end{cases}$$

Multiplication	Division
Even fn. $\times$ Even fn. = Even fn.	$\frac{\text{Even fn.}}{\text{Even fn.}} = \text{Even fn.}$
Odd fn. $\times$ Odd fn. = Even fn.	$\frac{\text{Odd fn.}}{\text{Odd fn.}} = \text{Even fn.}$
Even fn. $\times$ Odd fn. = Odd fn.	$\frac{\text{Even fn.}}{\text{Odd fn.}} = \text{Odd fn.}$
Odd fn. $\times$ Even fn. = Odd fn.	$\frac{\text{Odd fn.}}{\text{Even fn.}} = \text{Odd fn.}$

Fourier Transform Pair

The Fourier transform of $f(x)$ is defined by

$$F[f(x)] = F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

The inverse Fourier transform of $F(s)$ is defined by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$$

The Fourier transforms $F(s)$ of $f(x)$ and the inverse Fourier transform $f(x) = F^{-1}\{F(s)\}$ are jointly called Fourier transform pair.

Inversion Formula

If $f(x)$ is discontinuous at $x = x_0$, then the inversion formula can be

modified as $\frac{f(x_0+) + f(x_0-)}{2} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$ where $f(x_0+)$ and

$f(x_0-)$ denotes the right hand & left hand limits of $f(x)$ as $x \rightarrow x_0$.

Properties of Fourier Transform

Property 1 : Linear Property

If $F\{f(x)\} = F(s)$ and $F\{g(x)\} = G(s)$ then

$$F\{af(x) + bg(x)\} = aF(s) + bG(s)$$

Proof:

$$\begin{aligned} F\{af(x) + bg(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \{af(x) + bg(x)\} e^{isx} dx \\ &= a \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \right] + b \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{isx} dx \right] \\ &= aF(s) + bG(s) \end{aligned}$$

Where $F(s)$ and $G(s)$ are the Fourier transforms of $f(x)$ and $g(x)$.

Property 2 : Shifting Property

If $F\{f(x)\} = F(s)$, then $F\{f(x-a)\} = e^{isa}F(s)$

Proof :

$$F\{f(x-a)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-a)e^{isx} dx$$

$$\text{Put } t = x - a \Rightarrow dt = dx$$

$$\begin{aligned} \therefore F\{f(x-a)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{is(t+a)} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{ist} e^{isa} dt \\ &= e^{isa} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{ist} dt \\ &= e^{isa} F(s) \end{aligned}$$

Property 3 : Change of Scale Property

If $F\{f(x)\} = F(s)$, then $F\{f(ax)\} = \frac{1}{|a|} F\left(\frac{s}{a}\right)$ where $a \neq 0$

Proof :

$$F\{f(ax)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax)e^{isx} dx$$

$$\text{Put } t = ax \Rightarrow dt = adx \Rightarrow dx = \frac{dt}{a}$$

$$\therefore F\{f(ax)\} = \frac{1}{a} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{it\left(\frac{s}{a}\right)} dt \right]$$

$$\text{By definition, } F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{ist} dt$$

$$\therefore F\{f(ax)\} = \frac{1}{a} F\left(\frac{s}{a}\right) \text{ for } a > 0 \text{ and } = -\frac{1}{a} F\left(\frac{s}{a}\right) \text{ for } a < 0$$

$$\Rightarrow F\{f(ax)\} = \frac{1}{|a|} F\left(\frac{s}{a}\right) \text{ where } a \neq 0$$

Property 4 :

If $F\{f(x)\} = F(s)$, then $F\{e^{iax}f(x)\} = F(s+a)$

Proof :

$$\begin{aligned} \text{By definition, } F\{e^{iax}f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{isx} e^{iax} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{i(s+a)x} dx \end{aligned}$$

$$\text{But, } F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{isx} dx$$

$$F\{e^{iax}f(x)\} = F(s+a)$$

Property 5 : Modulation Property

If $F\{f(x)\} = F(s)$, then $F\{f(x)\cos ax\} = \frac{1}{2}[F(s-a) + F(s+a)]$

Proof :

$$\text{The Fourier transform of } f(x) \text{ is } F\{f(x)\} = F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{isx} dx$$

$$\therefore F\{f(x)\cos ax\} = F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\cos ax e^{isx} dx$$

$$\text{But, } \cos ax = \frac{e^{iax} + e^{-iax}}{2}$$

$$\begin{aligned} \therefore F\{f(x)\cos ax\} &= \frac{1}{2} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\{e^{iax} + e^{-iax}\}e^{isx} dx \right] \\ &= \frac{1}{2} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{i(s+a)x} dx \right] + \frac{1}{2} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{i(s-a)x} dx \right] \\ &= \frac{1}{2} [F(s+a) + F(s-a)] \end{aligned}$$

Property 6 :

If $F\{f(x)\} = F(s)$, then $F\{x^n f(x)\} = (-i)^n \frac{d^n}{ds^n} \{F(s)\}$

Proof :

The Fourier transform of $f(x)$ is $F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$

By differentiating $F(s)$ w.r.to s under the integral sign,

$$\begin{aligned} \frac{d}{ds} F(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \frac{\partial}{\partial s} \{e^{isx}\} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} (ix) dx \\ &= (i) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \{xf(x)\} e^{isx} dx \right] \\ \therefore (-i)F'(s) &= F\{xf(x)\} \end{aligned}$$

$$\text{Also } F\{x^2 f(x)\} = (-i) \frac{d}{ds} F\{xf(x)\}$$

$$\begin{aligned} &= (-i) \frac{d}{ds} [(-i)F'(s)] \\ &= (-i)^2 \frac{d^2}{ds^2} F(s) \end{aligned}$$

$$\text{In general, } F\{x^n f(x)\} = (-i)^n \frac{d^n}{ds^n} F(s)$$

Convolution of Two Functions

The Convolution of $f(x)$ and $g(x)$ is defined by

$$(f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t) dt$$

Convolution Theorem

If $F\{f(x)\} = F(s)$ and $F\{g(x)\} = G(s)$, then $F\{(f * g)(x)\} = F(s)G(s)$ where

$$(f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t) dt, \text{ the convolution of } f(x) \text{ and } g(x).$$

Proof :

$$\text{By definition, } (f * g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t) dt$$

The Fourier Transform of $f * g$ is

$$\begin{aligned} F\{f * g\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t) dt \right] e^{isx} dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) g(x-t) e^{isx} dt dx \end{aligned}$$

Changing the order of integration in R.H.S integral,

$$F\{f * g\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left[\int_{-\infty}^{\infty} g(x-t) e^{isx} dx \right] dt$$

$$\text{Put } x-t = u \Rightarrow x = u+t$$

Then $dx = du$ and u varies from $-\infty$ to ∞ .

$$\begin{aligned} \therefore F\{f * g\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left[\int_{-\infty}^{\infty} g(u) e^{is(u+t)} du \right] dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{ist} \left[\int_{-\infty}^{\infty} g(u) e^{isu} du \right] dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(u) e^{isu} du \right] dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} G(s) dt \left\{ \because G(s) = F\{g(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{isx} dx \right\} \\ \therefore F\{f * g\} &= \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{ist} dt \right] G(s) \\ &= F(s) G(s) \left\{ \because F(s) = F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \right\} \end{aligned}$$

$$\text{Hence } F\{f * g\} = F(s)G(s)$$

Note : By Convolution Theorem, $F\{f * g\} = F(s)G(s)$

$$\Rightarrow F^{-1}\{F(s)G(s)\} = \{f * g\}(x)$$

$$\text{i. e. } F^{-1}\{F(s)G(s)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)g(x-t)dt$$

Parseval's Identity

If $F(s)$ is the complex Fourier transform of $f(x)$, then

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$$

Proof :

By convolution theorem $F^{-1}\{F(s)G(s)\} = f * g$ where $f * g$ is the convolution of $f(x)$ and $g(x)$.

$$\therefore F^{-1}\{F(s)G(s)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)g(x-t)dt$$

$$\text{i. e. } \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)g(x-t)dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s)G(s)e^{-isx}ds$$

Put $x = 0$ in the above result,

$$\int_{-\infty}^{\infty} f(t)g(-t)dt = \int_{-\infty}^{\infty} F(s)G(s)ds \quad \text{--- (1)}$$

Let $g(t) = \overline{f(-t)} \therefore g(-t) = \overline{f(t)}$

$$\begin{aligned} G(s) &= F\{g(t)\} = F\{\overline{f(-t)}\} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ist} \overline{f(-t)} dt \end{aligned}$$

Put $t = -v$ in R.H.S integral

$$\begin{aligned} \therefore G(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overline{f(v)} e^{-isv} dv \\ &= \overline{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(v) e^{isv} dv} = \overline{F(s)} \end{aligned}$$

Where $\overline{F(s)}$ is the complex conjugate of $F(s)$.

$$\text{From (1), } \int_{-\infty}^{\infty} f(t)\overline{f(t)}dt = \int_{-\infty}^{\infty} F(s)\overline{F(s)}ds$$

$$\text{i. e. } \int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} |F(s)|^2 ds$$

$$\text{Hence } \int_{-\infty}^{\infty} |f(t)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$$

Illustrative Examples

1. Find the Fourier transform of $f(x) = \begin{cases} x & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$

Solution: The Fourier transform of $f(x)$ is

$$\begin{aligned} F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a x(\cos sx + i \sin sx) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a x \cos sx dx + \frac{i}{\sqrt{2\pi}} \int_{-a}^a x \sin sx dx \\ &= 0 + \frac{2i}{\sqrt{2\pi}} \int_0^a x \sin sx dx \\ &= \frac{\sqrt{2}\sqrt{2}i}{\sqrt{2\pi}} \left[x \left(-\frac{\cos sx}{s} \right) - \left(-\frac{\sin sx}{s^2} \right) \right]_0^a = i \sqrt{\frac{2}{\pi}} \left[-\frac{x \cos sx}{s} + \frac{\sin sx}{s^2} \right]_0^a \\ &= i \sqrt{\frac{2}{\pi}} \left[\left(-\frac{a \cos sa}{s} + \frac{\sin sa}{s^2} \right) - (0 + 0) \right] \\ &= i \sqrt{\frac{2}{\pi}} \left[\frac{\sin sa}{s^2} - \frac{a \cos sa}{s} \right] = i \sqrt{\frac{2}{\pi}} \left[\frac{\sin sa - a \cos sa}{s^2} \right] \\ F(s) &= i \sqrt{\frac{2}{\pi}} \left[\frac{\sin sa - a \cos sa}{s^2} \right] \end{aligned}$$

2. Show that the Fourier transform of $f(x) = \begin{cases} |x| & \text{for } |x| < a \\ 0 & \text{for } |x| > a, a > 0 \end{cases}$

$$\text{is } \sqrt{\frac{2}{\pi}} \left[\frac{s \cos sa + \cos sa - 1}{s^2} \right]$$

Solution: Given $|x| = \begin{cases} -x & \text{for } x < 0 \\ x & \text{for } x > 0 \end{cases}$

The Fourier transform of $f(x)$ is

$$\begin{aligned} F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a |x| (\cos sx + i \sin sx) dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\int_{-a}^a |x| \cos sx dx + i \int_{-a}^a |x| \sin sx dx \right] \\ &= \frac{2}{\sqrt{2\pi}} \int_0^a x \cos sx dx + 0 \\ &= \frac{\sqrt{2}\sqrt{2}}{\sqrt{2\pi}} \left[x \left(\frac{\sin sx}{s} \right) + \frac{\cos sx}{s^2} \right]_0^a \\ &= \sqrt{\frac{2}{\pi}} \left[\left(a \left(\frac{\sin sa}{s} \right) + \frac{\cos sa}{s^2} \right) - \left(0 - \frac{1}{s^2} \right) \right] = \sqrt{\frac{2}{\pi}} \left[\frac{s a \cos sa + \cos sa - 1}{s^2} \right] \end{aligned}$$

3. Find the Fourier transform of $f(x) = \begin{cases} 1 & ; |x| < 1 \\ 0 & ; |x| > 1 \end{cases}$. Hence deduce that

$$(i) \int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda = \frac{\pi}{2} \quad (ii) \int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

Solution: Given $f(x) = \begin{cases} 1 & ; |x| < 1 \\ 0 & ; |x| > 1 \end{cases}$

The Fourier transform of $f(x)$ is

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

$$\begin{aligned} F(s) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1) e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (\cos sx + i \sin sx) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 \cos sx dx + \frac{i}{\sqrt{2\pi}} \int_{-1}^1 \sin sx dx \\ &= \frac{2}{\sqrt{2\pi}} \int_0^1 \cos sx dx + 0 \\ &= \frac{\sqrt{2}\sqrt{2}}{\sqrt{2\pi}} \int_0^1 \cos sx dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^1 \cos sx dx \\ &= \sqrt{\frac{2}{\pi}} \left[\left(\frac{\sin sx}{s} \right) \right]_0^1 \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{\sin s}{s} - 0 \right) \\ F(s) &= \sqrt{\frac{2}{\pi}} \left[\frac{\sin s}{s} \right] \end{aligned}$$

$$(i) \text{ To Prove } \int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda = \frac{\pi}{2}$$

$$\text{By inversion formula } f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi}} \left[\frac{\sin s}{s} \right] e^{-isx} ds$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} \left[\frac{\sin s}{s} \right] (\cos sx - i \sin sx) ds \\
&= \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s}{s} \right] (\cos sx - i \sin sx) ds \\
&= \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s}{s} \right] (\cos sx) ds - \frac{i}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s}{s} \right] (\sin sx) ds \\
f(x) &= \frac{2}{\pi} \int_0^{\infty} \left[\frac{\sin s}{s} \right] (\cos sx) ds + 0 \\
\int_0^{\infty} \left[\frac{\sin s}{s} \right] (\cos sx) ds &= \frac{\pi}{2} f(x) \quad \text{--- (1)}
\end{aligned}$$

Put $x = 0$ in (1)

$$\int_0^{\infty} \left[\frac{\sin s}{s} \right] ds = \frac{\pi}{2} f(0), \text{ where } f(x) = \begin{cases} 1 & ; |x| < 1 \\ 0 & ; |x| > 1 \end{cases}, \text{ We have } f(0) = 1$$

$$\int_0^{\infty} \left[\frac{\sin s}{s} \right] ds = \frac{\pi}{2}$$

$$\text{Changing } s \text{ to } \lambda \int_0^{\infty} \left[\frac{\sin \lambda}{\lambda} \right] d\lambda = \frac{\pi}{2}$$

$$\text{(ii) To Prove } \int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

Using Parseval's identity

$$\int_{-\infty}^{\infty} |F(s)|^2 ds = \int_{-\infty}^{\infty} |f(x)|^2 dx$$

$$\int_{-\infty}^{\infty} \left(\sqrt{\frac{2}{\pi}} \frac{\sin s}{s} \right)^2 ds = \int_{-1}^1 (1)^2 dx, \quad \because f(x) = \begin{cases} 1 & ; -1 < x < 1 \\ 0 & ; x > 1 \end{cases}$$

$$\frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 s}{s^2} ds = (x) \Big|_{-1}^1 = 1 + 1 = 2$$

$$\int_{-\infty}^{\infty} \left(\frac{\sin s}{s} \right)^2 ds = \pi$$

$$2 \int_0^{\infty} \left(\frac{\sin s}{s} \right)^2 ds = \pi$$

$$\int_0^{\infty} \left(\frac{\sin s}{s} \right)^2 ds = \frac{\pi}{2}$$

Changing s to t , We have

$$\int_0^{\infty} \left(\frac{\sin t}{t} \right)^2 dt = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

4. Find the Fourier Transform of

$$f(x) = \begin{cases} a - |x| & ; |x| < a \\ 0 & ; |x| > a, a > 0 \end{cases} \text{ Hence deduce that (i) } \int_0^{\infty} \left(\frac{\sin t}{t} \right)^2 dt = \frac{\pi}{2}$$

$$\text{(ii) } \int_0^{\infty} \frac{\sin^4 t}{t^4} dt = \frac{\pi}{3}$$

Solution: The Fourier transform of $f(x)$ is

$$\begin{aligned}
F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-a}^a (a - |x|) (\cos sx + i \sin sx) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-a}^a (a - |x|) \cos sxdx + \cancel{\frac{i}{\sqrt{2\pi}} \int_{-a}^a (a - |x|) \sin sxdx} \\
&= \frac{2}{\sqrt{2\pi}} \int_0^a (a - x) \cos sxdx + 0 \\
&= \frac{\sqrt{2}\sqrt{2}}{\sqrt{2\pi}} \int_0^a (a - x) \cos sxdx = \sqrt{\frac{2}{\pi}} \int_0^a (a - x) \cos sxdx \\
&= \sqrt{\frac{2}{\pi}} \left[\left\{ (a - x) \left(\frac{\sin sx}{s} \right) \right\} - \left\{ (-1) \left(\frac{-\cos sx}{s^2} \right) \right\} \right]_0^a
\end{aligned}$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \left[0 - \frac{\cos sa}{s^2} \right] - \left[0 - \frac{1}{s^2} \right] \right\}$$

$$F(s) = \sqrt{\frac{2}{\pi}} \left[\frac{1 - \cos sa}{s^2} \right]$$

(i) To Prove $\int_0^\infty \left(\frac{\sin t}{t} \right)^2 dt = \frac{\pi}{2}$

By inversion formula, $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty F(s) e^{-isx} ds$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \sqrt{\frac{2}{\pi}} \left[\frac{1 - \cos sa}{s^2} \right] e^{-isx} ds$$

$$= \frac{1}{\pi} \int_{-\infty}^\infty \left[\frac{1 - \cos sa}{s^2} \right] (\cos sx - i \sin sx) ds$$

$$= \frac{1}{\pi} \int_{-\infty}^\infty \left[\frac{1 - \cos sa}{s^2} \right] \cos sxdx - \frac{i}{\pi} \int_{-\infty}^\infty \left[\frac{1 - \cos sa}{s^2} \right] \sin sxdx$$

$$f(x) = \frac{2}{\pi} \int_0^\infty \left[\frac{1 - \cos sa}{s^2} \right] \cos sxdx + 0$$

$$= \frac{2}{\pi} \int_0^\infty \left[\frac{2 \sin^2 \left(\frac{sa}{2} \right)}{s^2} \right] \cos sxdx$$

$$f(x) = \frac{4}{\pi} \int_0^\infty \left[\frac{\sin^2 \left(\frac{sa}{2} \right)}{s^2} \right] \cos sxdx$$

Put $x = 0$, we get

$$f(0) = \frac{4}{\pi} \int_0^\infty \left[\frac{\sin^2 \left(\frac{sa}{2} \right)}{s^2} \right] \cos 0 ds$$

$$a = \frac{4}{\pi} \int_0^\infty \left[\frac{\sin^2 \left(\frac{sa}{2} \right)}{s^2} \right] ds$$

Since $f(x) = \begin{cases} a - |x|; & -a < x < a \\ 0; & \text{otherwise} \end{cases}$; we have $f(0) = a - 0 = a$

$$\text{Put } \left(\frac{sa}{2} \right) = t \Rightarrow s = \left(\frac{2t}{a} \right) \Rightarrow ds = \frac{2dt}{a}$$

$$\Rightarrow \frac{4}{\pi} \int_0^\infty \left[\frac{\sin^2(t)}{\left(\frac{2t}{a} \right)^2} \right] \left(\frac{2}{a} \right) dt = a$$

$$\Rightarrow \frac{4}{\pi} \int_0^\infty [\sin^2(t)] \left(\frac{a^2}{4t^2} \right) \left(\frac{2}{a} \right) dt = a$$

$$\Rightarrow \frac{2a}{\pi} \int_0^\infty \left[\frac{\sin^2(t)}{t^2} \right] dt = a$$

$$\Rightarrow \int_0^\infty \left[\frac{\sin^2(t)}{t^2} \right] dt = \frac{\pi}{2}$$

(ii) To Prove $\int_0^\infty \frac{\sin^4 t}{t^4} dt = \frac{\pi}{3}$

By using Parseval's identity, $\int_{-\infty}^\infty |F(s)|^2 ds = \int_{-\infty}^\infty |f(x)|^2 dx$

$$\int_{-\infty}^\infty \frac{2}{\pi} \left(\frac{1 - \cos sa}{s^2} \right)^2 ds = \int_{-a}^a (a - |x|)^2 dx$$

$$\frac{2}{\pi} \int_{-\infty}^\infty \left(\frac{2 \sin^2 \left(\frac{sa}{2} \right)}{s^2} \right)^2 ds = \int_{-a}^a (a - |x|)^2 dx$$

$$= 2 \int_0^a (a - |x|)^2 dx$$

$$\Rightarrow \frac{4}{\pi} \int_0^\infty \left(\frac{2 \sin^2 \left(\frac{sa}{2} \right)}{s^2} \right)^2 ds = 2 \int_0^a (a - x)^2 dx$$

$$= 2 \left[-\frac{(a - x)^3}{3} \right]_0^a$$

$$= 2 \left[0 - \left(-\frac{(a - 0)^3}{3} \right) \right] = \frac{2a^3}{3}$$

$$\int_0^{\infty} \left(\frac{2\sin^2\left(\frac{sa}{2}\right)}{s^2} \right)^2 ds = \frac{2a^3}{3}$$

$$\text{Put } \left(\frac{sa}{2}\right) = t \Rightarrow s = \left(\frac{2t}{a}\right) \Rightarrow ds = \frac{2dt}{a}$$

$$\Rightarrow \frac{4}{\pi} \int_0^{\infty} \left(\frac{2\sin^2(t)}{\left(\frac{2t}{a}\right)^2} \right)^2 \frac{2}{a} dt = \frac{2a^3}{3}$$

$$\Rightarrow \frac{4}{\pi} \int_0^{\infty} \left(\frac{4\sin^4(t)}{\frac{16t^4}{a^4}} \right) \frac{2}{a} dt = \frac{2a^3}{3}$$

$$\Rightarrow \frac{16}{\pi} \int_0^{\infty} \sin^4 t \frac{a^4}{16t^4} \frac{2}{a} dt = \frac{2a^3}{3}$$

$$\Rightarrow \frac{2a^3}{\pi} \int_0^{\infty} \frac{\sin^4 t}{t^4} dt = \frac{2a^3}{3}$$

$$\Rightarrow \int_0^{\infty} \frac{\sin^4 t}{t^4} dt = \frac{2a^3}{3} \frac{\pi}{2a^3} = \frac{\pi}{3}$$

$$\int_0^{\infty} \frac{\sin^4 t}{t^4} dt = \frac{\pi}{3}$$

5. Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2; & |x| < 1 \\ 0 & ; |x| > 1 \end{cases}$ and hence evaluate

$$(i) \int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right) dt \quad (ii) \int_0^{\infty} \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$$

Solution : The Fourier transform of $f(x)$ is

$$\begin{aligned} F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1 - x^2) e^{isx} dx \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1 - x^2) (\cos sx + i \sin sx) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1 - x^2) \cos sxdx + \frac{i}{\sqrt{2\pi}} \int_{-1}^1 (1 - x^2) \sin sxdx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 (1 - x^2) \cos sxdx + 0$$

$$= \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}\sqrt{\pi}} \int_0^1 (1 - x^2) \cos sxdx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 (1 - x^2) \cos sxdx$$

$$= \sqrt{\frac{2}{\pi}} \left\{ (1 - x^2) \left(\frac{\sin sx}{s} \right) - (-2x) \left(\frac{-\cos sx}{s^2} \right) - 2 \left(\frac{-\sin sx}{s^3} \right) \right\}_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \left[0 - \frac{2\cos s}{s^2} + \frac{2\sin s}{s^3} \right] - 0 \right\}$$

$$F(s) = 2 \sqrt{\frac{2}{\pi}} \left[\frac{\sin s - s \cos s}{s^3} \right]$$

$$(i) \text{ To Find } \int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right) dt$$

By Fourier inversion formula

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 2 \sqrt{\frac{2}{\pi}} \left[\frac{\sin s - s \cos s}{s^3} \right] e^{-isx} ds$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] e^{-isx} ds$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] (\cos sx - i \sin sx) ds$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] \cos sx ds - \frac{2i}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] \sin sx ds$$

$$f(x) = \frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] \cos sx ds + 0 \quad \text{--- (1)}$$

Put $x = 0$ in (1), we get,

$$\frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] ds = f(0) = 1$$

Where $f(x) = \begin{cases} 1 - x^2; & -1 < x < 1 \\ 0; & \text{otherwise} \end{cases}$; We have $f(0) = 1 - (0)^2 = 1$

Put $s = t \Rightarrow ds = dt$

$$\Rightarrow \frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin t - t \cos t}{t^3} \right] dt = 1 \Rightarrow \int_0^{\infty} \left[\frac{\sin t - t \cos t}{t^3} \right] dt = \frac{\pi}{4}$$

$$(ii) \text{ To find } \int_0^{\infty} \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$$

Put $x = \frac{1}{2}$ in (1) we get,

$$f\left(\frac{1}{2}\right) = \frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] \cos \frac{s}{2} ds, \text{ where } f(x) = \begin{cases} 1 - x^2; & -1 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

$$\text{We have } f\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{3}{4} = \frac{4}{\pi} \int_0^{\infty} \left[\frac{\sin s - s \cos s}{s^3} \right] \cos \frac{s}{2} ds$$

$$\frac{3\pi}{16} = - \int_0^{\infty} \left[\frac{s \cos s - \sin s}{s^3} \right] \cos \frac{s}{2} ds$$

Changing sto x , we get,

$$\frac{-3\pi}{16} = \int_0^{\infty} \left[\frac{x \cos x - \sin x}{x^3} \right] \cos \frac{x}{2} dx$$

6. Find the Fourier transform of $f(x) = \begin{cases} 4 - x^2; & |x| < 2 \\ 0 & ; |x| > 2 \end{cases}$ and hence evaluate

$$\int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right)^2 dt = \frac{\pi}{15}$$

Solution: The Fourier transform of $f(x)$ is

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-2}^2 (4 - x^2) e^{isx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-2}^2 (4 - x^2) (\cos sx + i \sin sx) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-2}^2 (4 - x^2) \cos sx dx + \frac{i}{\sqrt{2\pi}} \int_{-2}^2 (4 - x^2) \sin sx dx$$

$$F\{f(x)\} = \frac{2}{\sqrt{2\pi}} \int_0^2 (4 - x^2) \cos sx dx + 0$$

$$= \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}\sqrt{\pi}} \int_0^2 (4 - x^2) \cos sx dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^2 (4 - x^2) \cos sx dx$$

$$= \sqrt{\frac{2}{\pi}} \left\{ (4 - x^2) \left(\frac{\sin sx}{s} \right) - (-2x) \left(\frac{-\cos sx}{s^2} \right) + (-2) \left(\frac{-\sin sx}{s^3} \right) \right\}_0^2$$

$$= \sqrt{\frac{2}{\pi}} \left\{ (0) - \left(\frac{4 \cos 2s}{s^2} \right) + \left(\frac{2 \sin 2s}{s^3} \right) - 0 \right\}$$

$$= 2 \sqrt{\frac{2}{\pi}} \left\{ \left(\frac{\sin 2s}{s^3} \right) - \left(\frac{2 \cos 2s}{s^2} \right) \right\}$$

$$F(s) = 2 \sqrt{\frac{2}{\pi}} \left\{ \left(\frac{\sin 2s}{s^3} \right) - \left(\frac{2 \cos 2s}{s^2} \right) \right\}$$

$$F(s) = 2 \sqrt{\frac{2}{\pi}} \left\{ \left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right) \right\}$$

By Parseval's identity, $\int_{-\infty}^{\infty} |F(s)|^2 ds = \int_{-\infty}^{\infty} |f(x)|^2 dx$

$$\int_{-\infty}^{\infty} \left\{ 2 \sqrt{\frac{2}{\pi}} \left[\left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right) \right] \right\}^2 ds = \int_{-2}^2 (4 - x^2)^2 dx$$

$$2 \int_0^{\infty} 4 \cdot \frac{2}{\pi} \left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right)^2 ds = 2 \int_0^2 (4 - x^2)^2 dx$$

$$\frac{16}{\pi} \int_0^{\infty} \left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right)^2 ds = 2 \int_0^2 (4 - x^2)^2 dx$$

$$= 2 \int_0^2 (16 + x^4 - 8x^2) dx$$

$$= 2 \left(16x + \frac{x^5}{5} - \frac{8x^3}{3} \right)_0^2$$

$$= 2 \left\{ 32 + \frac{32}{5} - \frac{64}{3} \right\} = 2(32) \left\{ 1 + \frac{1}{5} - \frac{2}{3} \right\}$$

$$\Rightarrow \frac{16}{\pi} \int_0^{\infty} \left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right)^2 ds = 64 \left\{ \frac{8}{15} \right\}$$

$$\frac{1}{\pi} \int_0^{\infty} \left(\frac{\sin 2s - 2s \cos 2s}{s^3} \right)^2 ds = \frac{32}{15}$$

$$\text{Put } 2s = t \Rightarrow s = \frac{t}{2} \Rightarrow ds = \frac{dt}{2}$$

$$\frac{1}{\pi} \int_0^{\infty} \left(\frac{\sin t - t \cos t}{\left(\frac{t}{2}\right)^3} \right)^2 \frac{dt}{2} = \frac{32}{15}$$

$$\frac{1}{\pi} \int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right)^2 64 \cdot \frac{dt}{2} = \frac{32}{15}$$

$$\frac{1}{\pi} \int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right)^2 \cdot 32 \cdot dt = \frac{32}{15}$$

$$\int_0^{\infty} \left(\frac{\sin t - t \cos t}{t^3} \right)^2 dt = \frac{\pi}{15}$$

Self-reciprocal

If the Fourier transform of $f(x)$ is $f(s)$, then $f(x)$ is said to be self-reciprocal under Fourier transform.

7. Find the Fourier transform of $e^{-a^2 x^2}$, $a > 0$. Hence Show that $e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier transform.

Solution: Let $f(x) = e^{-a^2 x^2}$, $a > 0$

The Fourier transform of $f(x)$ is

$$\begin{aligned} F\{f(x)\} &= F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a^2 x^2} e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a^2 x^2 + isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(a^2 x^2 - isx)} dx \end{aligned} \quad \text{--- (1)}$$

$$\{\because a^2 - 2ab + b^2 = (a - b)^2\}$$

$$\begin{aligned} -(a^2 x^2 - isx) &= -((ax)^2 - isx) = -\left[(ax)^2 - \frac{2isax}{2a} + \left(\frac{is}{2a}\right)^2 - \left(\frac{is}{2a}\right)^2 \right] \\ &= -\left[\left(ax - \frac{is}{2a}\right)^2 - \left(\frac{is}{2a}\right)^2 \right] \\ &= \left[-\left(ax - \frac{is}{2a}\right)^2 + \left(\frac{is}{2a}\right)^2 \right] \end{aligned}$$

(1) becomes,

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left[-(ax - \frac{is}{2a})^2 + \left(\frac{is}{2a}\right)^2\right]} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left[-(ax - \frac{is}{2a})^2\right]} e^{\left(\frac{is}{2a}\right)^2} dx \\
&= \frac{1}{\sqrt{2\pi}} e^{\left(\frac{is}{2a}\right)^2} \int_{-\infty}^{\infty} e^{\left[-(ax - \frac{is}{2a})^2\right]} dx \quad \text{--- (2)}
\end{aligned}$$

$$\text{Put } ax - \frac{is}{2a} = u \Rightarrow adx = du \Rightarrow dx = \frac{du}{a}$$

Then (2) becomes,

$$\begin{aligned}
F(s) &= \frac{1}{\sqrt{2\pi}} e^{\left(\frac{is}{2a}\right)^2} \int_{-\infty}^{\infty} e^{-u^2} \frac{1}{a} du \\
&= \frac{1}{a\sqrt{2\pi}} e^{\frac{-s^2}{4a^2}} \int_{-\infty}^{\infty} e^{-u^2} du \\
&= \frac{2}{a\sqrt{2\pi}} e^{\frac{-s^2}{4a^2}} \int_0^{\infty} e^{-u^2} du \quad (\because e^{-u^2} \text{ is even})
\end{aligned}$$

$$\text{Put } t = u^2 \Rightarrow dt = 2u du \text{ and } u = \sqrt{t} \Rightarrow du = \frac{dt}{2u} = \frac{dt}{2\sqrt{t}}$$

$$\begin{aligned}
F(s) &= \frac{2e^{\frac{-s^2}{4a^2}}}{a\sqrt{2\pi}} \int_0^{\infty} e^{-t} \frac{dt}{2\sqrt{t}} \\
&= \frac{2e^{\frac{-s^2}{4a^2}}}{2a\sqrt{2\pi}} \int_0^{\infty} e^{-t} t^{-\frac{1}{2}} dt \\
&= \frac{e^{\frac{-s^2}{4a^2}}}{a\sqrt{2\pi}} \int_0^{\infty} e^{-t} t^{\frac{1}{2}-1} dt \\
&= \frac{e^{\frac{-s^2}{4a^2}}}{a\sqrt{2\pi}} \Gamma\left(\frac{1}{2}\right) \left\{ \because \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \Rightarrow \Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} e^{-t} t^{\frac{1}{2}-1} dt \right\}
\end{aligned}$$

$$\begin{aligned}
&= \frac{e^{\frac{-s^2}{4a^2}} \sqrt{\pi}}{a\sqrt{2\pi}} \quad (\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}) \\
&= \frac{e^{\frac{-s^2}{4a^2}}}{a\sqrt{2}}
\end{aligned}$$

Put $a = \frac{1}{\sqrt{2}}$ in above equation, we get

$$= \frac{1}{\left(\frac{1}{\sqrt{2}}\right)\sqrt{2}} e^{\frac{-s^2}{4\left(\frac{1}{\sqrt{2}}\right)^2}} = e^{\frac{-s^2}{2}}$$

Since $F\left(e^{\frac{-x^2}{2}}\right) = e^{\frac{-s^2}{2}}$, the function $f(x) = e^{\frac{-x^2}{2}}$ is self reciprocal under Fourier transform.

8. Prove that the Fourier transform of $e^{\frac{-x^2}{2}}$ is $e^{\frac{-s^2}{2}}$ and deduce that

$$F\left(xe^{\frac{-x^2}{2}}\right) = (is) e^{\frac{-s^2}{2}}$$

Solution: Let $f(x) = e^{\frac{-x^2}{2}}$

The Fourier transform of $f(x)$ is

$$\begin{aligned}
F\{f(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \\
F\left(e^{\frac{-x^2}{2}}\right) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-x^2}{2}} e^{isx} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-x^2}{2} + isx} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - 2isx)} dx \quad \text{--- (1)}
\end{aligned}$$

$$\text{Since } a^2 - 2ab + b^2 = (a - b)^2$$

$$\begin{aligned}
-\frac{1}{2}(x^2 - 2isx) &= -\frac{1}{2}[(x)^2 - 2isx + (is)^2 - (is)^2] \\
&= -\frac{1}{2}[(x - is)^2 + s^2]
\end{aligned}$$

$$= -\frac{1}{2}(x - is)^2 - \frac{s^2}{2}$$

$$\begin{aligned}\text{Now (1) becomes } F\left(e^{-\frac{x^2}{2}}\right) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-is)^2 - \frac{s^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-is)^2} e^{-\frac{s^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{s^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-is)^2} dx\end{aligned}$$

Put $x - is = t \Rightarrow dx = dt$

$$F(s) = \frac{1}{\sqrt{2\pi}} e^{-\frac{s^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt \quad \text{--- (2)}$$

$$\text{Consider } \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt$$

Since $f(t)$ is even function

$$\int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = 2 \int_0^{\infty} e^{-\frac{t^2}{2}} dt \quad \text{--- (3)}$$

$$\text{Put } \frac{t^2}{2} = u \Rightarrow t^2 = 2u \Rightarrow 2tdt = 2du \text{ and } t = \sqrt{2}\sqrt{u}$$

$$\Rightarrow dt = \frac{du}{t} = \frac{du}{\sqrt{2}\sqrt{u}}$$

$$\begin{aligned}(3) \Rightarrow 2 \int_0^{\infty} e^{-u} \frac{du}{\sqrt{2}\sqrt{u}} &= \sqrt{2} \int_0^{\infty} e^{-u} u^{\frac{1}{2}-1} du \\ &= \sqrt{2} \Gamma\left(\frac{1}{2}\right) = \sqrt{2\pi} \quad (\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi})\end{aligned}$$

$$\left\{ \text{We know that } \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \Rightarrow \Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} e^{-u} u^{\frac{1}{2}-1} du \right\}$$

Then (2) becomes,

$$F(s) = \frac{1}{\sqrt{2\pi}} e^{-\frac{s^2}{2}} \sqrt{2\pi} = e^{-\frac{s^2}{2}}$$

$$F\left(e^{-\frac{x^2}{2}}\right) = e^{-\frac{s^2}{2}} \Rightarrow e^{-\frac{x^2}{2}} \text{ is self reciprocal under Fourier transform}$$

$$\text{Next to find } F\left(xe^{-\frac{x^2}{2}}\right)$$

$$\text{By using property, } F(xf(x)) = (-i) \frac{d}{ds} [F(s)]$$

$$\text{Where } F(s) = F\{f(x)\}$$

$$\begin{aligned}F\left(xe^{-\frac{x^2}{2}}\right) &= (-i) \frac{d}{ds} \left[e^{-\frac{s^2}{2}}\right] \\ &= (-i) e^{-\frac{s^2}{2}} \left[-\frac{2s}{2}\right] \\ &= (-i)(-s) e^{-\frac{s^2}{2}} \Rightarrow F\left(xe^{-\frac{x^2}{2}}\right) = (is) e^{-\frac{s^2}{2}}\end{aligned}$$

9. Find the Fourier transform of $e^{-a|x|}$ if $a > 0$. Deduce that

$$\int_0^{\infty} \frac{\cos x dx}{(x^2 + a^2)^2} = \frac{\pi}{2a} e^{-|x|}. \text{ Hence find Fourier transform of } xe^{-a|x|}$$

Solution:

$$\text{Let } f(x) = e^{-a|x|}$$

The Fourier transform of $f(x)$ is

$$F\{f(x)\} = F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$$

$$\begin{aligned}F(e^{-a|x|}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} e^{isx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} (\cos sx + i \sin sx) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} \cos sx dx + \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} \sin sx dx \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-ax} \cos sx dx + 0\end{aligned}$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \cos sx dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \cos sx dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \cos sx dx$$

$$\left(\because \int e^{ax} \cos bx dx = \frac{e^{ax}}{(a^2 + b^2)} [a \cos bx + b \sin bx] \right)$$

Here $a = -a$ & $b = s$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-ax}}{(a^2 + s^2)} [-a \cos sx + s \sin sx] \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-\infty}}{(a^2 + s^2)} [-a \cos s\infty + s \sin s\infty] - \left[\frac{e^0}{(a^2 + s^2)} [-a \cos s0 + s \sin s0] \right] \right]$$

$$(\because e^{-\infty} = 0, e^0 = 1, \cos 0 = 1, \& \sin 0 = 0)$$

$$= \sqrt{\frac{2}{\pi}} \left[\{0\} - \left\{ \frac{1}{(a^2 + s^2)} [-a(1) + 0] \right\} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[-\frac{1}{(a^2 + s^2)} [-a] \right]$$

$$F(s) = \sqrt{\frac{2}{\pi}} \left[\frac{a}{(a^2 + s^2)} \right]$$

By Fourier inversion theorem,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi}} \left[\frac{a}{(a^2 + s^2)} \right] e^{-isx} ds$$

$$= \frac{a}{\pi} \int_{-\infty}^{\infty} \left[\frac{1}{(a^2 + s^2)} \right] (\cos sx - i \sin sx) ds$$

$$= \frac{a}{\pi} \int_{-\infty}^{\infty} \left[\frac{\cos sx}{(a^2 + s^2)} \right] ds - \frac{ai}{\pi} \int_{-\infty}^{\infty} \left[\frac{\sin sx}{(a^2 + s^2)} \right] ds$$

$$= \frac{2a}{\pi} \int_0^{\infty} \left[\frac{\cos sx}{(a^2 + s^2)} \right] ds + 0$$

$$\Rightarrow e^{-a|x|} = \frac{2a}{\pi} \int_0^{\infty} \left[\frac{\cos sx}{(a^2 + s^2)} \right] ds \quad (\because f(x) = e^{-a|x|})$$

$$\Rightarrow \frac{\pi}{2a} e^{-a|x|} = \int_0^{\infty} \left[\frac{\cos sx}{(a^2 + s^2)} \right] ds$$

Changing s to t

$$\Rightarrow \int_0^{\infty} \left[\frac{\cos tx}{(a^2 + t^2)} \right] dt = \frac{\pi}{2a} e^{-a|x|}$$

By using property, $F(xf(x)) = (-i) \frac{d}{ds} [F(s)]$

where $F(s) = F\{f(x)\}$

Taking $f(x) = e^{-a|x|}$, we have

$$F\{xe^{-a|x|}\} = \sqrt{\frac{2}{\pi}} (-i) \frac{d}{ds} \left[\frac{a}{(a^2 + s^2)} \right]$$

$$= \sqrt{\frac{2}{\pi}} (-i) \frac{(a^2 + s^2)0 - 2as}{(a^2 + s^2)^2} = \sqrt{\frac{2}{\pi}} \frac{2ias}{(a^2 + s^2)^2}$$

Problems for Practice

1. Find the complex Fourier transform of $f(x) = \begin{cases} x^2, & |x| < a \\ 0, & |x| > a \end{cases}$

$$\text{Ans : } \sqrt{\frac{2}{\pi} \left(\frac{(a^2 s^2 - 2) \sin sa + 2 a \cos sa}{s^3} \right)}$$

2. Find the Fourier transform of $f(x) = \begin{cases} xe^x, & x > 0 \\ 0, & x < 0 \end{cases}$

$$\text{Ans : } \frac{1}{\sqrt{2\pi}(1-is)^2}$$

3. Find the Fourier transform of $e^{-|x|}$. Using Parseval's identity, evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)^2} = \frac{\pi}{4}$$

FOURIER SINE & COSINE TRANSFORMS

Fourier sine transforms (FST)

The infinite Fourier sine transforms of $f(x)$ is defined by

$$F_s\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx$$

It is also denoted by $F_s(s)$

The inverse Fourier sine transform of $F_s(s)$ is defined by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(s) \sin sx \, ds$$

Fourier cosine transforms (FCT)

The infinite Fourier cosine transform of $f(x)$ is defined by

$$F_c\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$$

It is also denoted by $F_c(s)$

The inverse Fourier cosine transform of $F_c(s)$ is defined by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c(s) \cos sx \, ds$$

Properties of FCT and FST

Property 1: Linear Property

If $F_c\{f(x)\} = F_c(s)$ and $F_c\{g(x)\} = G_c(s)$ then

$$\begin{aligned} F_c\{af(x) + bg(x)\} &= aF_c(s) + bG_c(s) \\ &= aF_c\{f(x)\} + bF_c\{g(x)\} \end{aligned}$$

Property 2

If $F_c\{f(x)\} = F_c(s)$ is the F.C.T of $f(x)$, then

$$F_c\{f(x) \cos ax\} = \frac{1}{2} \{F_c(s+a) + F_c(s-a)\}$$

Property 3

If $F_s(s)$ is the F.S.T of (x) , then

$$F_c\{f(x) \sin ax\} = \frac{1}{2} \{F_s(s+a) - F_s(s-a)\}$$

Property 4

If $F_s(s)$ is the F.S.T of (x) , then

$$F_s\{f(x) \cos ax\} = \frac{1}{2} \{F_s(s+a) + F_s(s-a)\}$$

Property 5

If $F_c(s)$ is the F.C.T of $f(x)$, then

$$F_s\{f(x) \sin ax\} = \frac{1}{2} \{F_c(s-a) - F_c(s+a)\}$$

Property 6

If $F_c(s)$ is the F.C.T of $f(x)$ and $F_s(s)$ is F.S.T of $f(x)$, then

$$F_c\{f(ax)\} = \frac{1}{a} F_c\left(\frac{s}{a}\right) \text{ and } F_s\{f(ax)\} = \frac{1}{a} F_s\left(\frac{s}{a}\right)$$

Parseval's identity for Fourier sine transforms

If $F_s(s)$ & $G_s(s)$ are Fourier sine transforms of $f(x)$ & $g(x)$ respectively, then

$$\int_0^\infty F_s(s) G_s(s) ds = \int_0^\infty f(x) g(x) dx$$

Parseval's identity for Fourier cosine transform

If $F_c(s)$ & $G_c(s)$ are Fourier cosine transforms of $f(x)$ & $g(x)$ respectively, then

$$\int_0^\infty F_c(s) \cdot G_c(s) ds = \int_0^\infty f(x) g(x) dx$$

Result

If $F_c(s)$ are Fourier cosine transform of $f(x)$ then,

$$\int_0^\infty [F_c(s)]^2 ds = \int_0^\infty [f(x)]^2 dx \text{ and}$$

If $F_s(s)$ are Fourier sine transform of $f(x)$ then,

$$\int_0^\infty [F_s(s)]^2 ds = \int_0^\infty [f(x)]^2 dx$$

Illustrative Examples

1. Find the Fourier sine and cosine transform of $f(x) = \begin{cases} 1 & \text{for } 0 < x < a \\ 0 & \text{for } x > a \end{cases}$

Solution : The F. S. T of $f(x)$ is $F_s\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin sx \, dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^a 1 \cdot \sin sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^a \sin sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left(-\frac{\cos sx}{s} \right)_0^a$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{1 - \cos sa}{s} \right)$$

The F. C. T of $f(x)$ is $F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos sx \, dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^a \cos sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{\sin sx}{s} \right)_0^a = \sqrt{\frac{2}{\pi}} \frac{\sin sa}{s}$$

2. Obtain the Fourier Sine Transform of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$

Solution : The F. S. T of $f(x)$ is $F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin sx \, dx$

$$F_s(S) = \sqrt{\frac{2}{\pi}} \left[\int_0^1 x \sin sx \, dx + \int_1^2 (2 - x) \sin sx \, dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[x \left(-\frac{\cos sx}{s} \right) + \frac{\sin sx}{s^2} \right]_0^1 + \sqrt{\frac{2}{\pi}} \left[(2 - x) \left(-\frac{\cos sx}{s} \right) - \frac{\sin sx}{s^2} \right]_1^2$$

$$= \sqrt{\frac{2}{\pi}} \left[-\frac{\cos s}{s} + \frac{\sin s}{s^2} + \frac{\cos s}{s} - \frac{\sin 2s}{s^2} + \frac{\sin s}{s^2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{2 \sin s}{s^2} - \frac{\sin 2s}{s^2} \right]$$

$$= \sqrt{\frac{2}{\pi} \left[\frac{2 \sin s - 2 \sin s \cos s}{s^2} \right]}$$

$$= \sqrt{\frac{2}{\pi} \left[\frac{2 \sin s (1 - \cos s)}{s^2} \right]}$$

3. Find the Fourier cosine transform of $f(x) = \begin{cases} 1 - x^2, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Hence prove $\int_0^\infty \left(\frac{\sin x - x \cos x}{x^3} \right) \cos \left(\frac{x}{2} \right) dx = \frac{3\pi}{16}$

Solution: The F.C.T of $f(x)$ is

$$F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos sx \, dx = \sqrt{\frac{2}{\pi}} \int_0^\infty (1 - x^2) \cos sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left((1 - x^2) \left(\frac{\sin sx}{s} \right) - (-2x) \left(-\frac{\cos sx}{s^2} \right) + (-2) \left(-\frac{\sin sx}{s^3} \right) \right) \Big|_0^\infty$$

$$= \sqrt{\frac{2}{\pi}} \left(\left(0 - \frac{2 \cos s}{s^2} + \frac{2 \sin s}{s^3} \right) - (0) \right)$$

$$= \sqrt{\frac{2}{\pi}} \left(-\frac{2 \cos s}{s^2} + \frac{2 \sin s}{s^3} \right)$$

$$F_c(s) = 2 \sqrt{\frac{2}{\pi}} \left(\frac{\sin s - s \cos s}{s^3} \right)$$

By inversion Formula

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F_c(s) \cos sx \, ds$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty 2 \sqrt{\frac{2}{\pi}} \left(\frac{\sin s - s \cos s}{s^3} \right) \cos sx \, ds$$

$$= \frac{4}{\pi} \int_0^\infty \left(\frac{\sin s - s \cos s}{s^3} \right) \cos sx \, ds$$

Put $x = \frac{1}{2}$, We get

$$\frac{4}{\pi} \int_0^\infty \left(\frac{\sin s - s \cos s}{s^3} \right) \cos \frac{s}{2} \, ds = f\left(\frac{1}{2}\right) = \frac{3}{4}$$

Where $f(x) = \begin{cases} 1 - x^2, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$; We have $f\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$

$$\int_0^\infty \left(\frac{\sin s - s \cos s}{s^3} \right) \cos \frac{s}{2} \, ds = \frac{3\pi}{16}$$

4. Find the Fourier transform of e^{-x} . Hence show that

$$\int_0^\infty \frac{x \sin mx}{1 + x^2} \, dx = \frac{\pi e^{-m}}{2}, m > 0$$

Solution: The F.S.T of $f(x)$ is $F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin sx \, dx$

Here $f(x) = e^{-x}$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \sin sx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{e^{-x}}{(-1)^2 + s^2} (-\sin sx - s \cos sx) \right]_0^\infty$$

$$\left(\because \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right) \text{ Here } a = -1, b = s$$

$$= \sqrt{\frac{2}{\pi}} \left[0 - \frac{1}{1^2 + s^2} (0 - (-s)) \right]$$

$$F_s(s) = \sqrt{\frac{2}{\pi}} \left(\frac{s}{s^2 + 1} \right)$$

By inversion formula, $f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty F_s(s) \sin sx \, ds$

$$\begin{aligned} &= \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{\frac{2}{\pi}} \left(\frac{s}{s^2 + 1} \right) \sin sx \, ds \\ \Rightarrow f(x) &= \frac{2}{\pi} \int_0^\infty \frac{s \sin sx}{s^2 + 1} \, ds \\ \Rightarrow e^{-x} &= \frac{2}{\pi} \int_0^\infty \frac{s \sin sx}{s^2 + 1} \, ds \\ \Rightarrow \int_0^\infty \frac{s \sin sx}{s^2 + 1} \, ds &= \frac{\pi}{2} e^{-x} \end{aligned}$$

Changing x to m and s to x in the above result,

$$\int_0^\infty \frac{x \sin mx}{1 + x^2} \, dx = \frac{\pi e^{-m}}{2}$$

5. Find the Fourier transform of $e^{-a|x|}$ if $a > 0$. Deduce that

$$\int_0^\infty \frac{1}{(x^2 + a^2)^2} \, dx = \frac{\pi}{4a^3} \text{ if } a > 0.$$

Solution: The Fourier transform of $f(x)$ is

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty f(x) e^{isx} \, dx$$

Here $f(x) = e^{-a|x|}$

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-a|x|} e^{isx} \, dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-a|x|} (\cos sx + i \sin sx) \, dx \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-a|x|} \cos sx \, dx + \frac{i}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-a|x|} \sin sx \, dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-ax} \cos sx \, dx + 0$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cos sx \, dx$$

$$\left(\because \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{(a^2 + b^2)} [a \cos bx + b \sin bx] \right) \text{ Here } a = -a \text{ \& } b = s$$

$$= \left(\frac{e^{-ax}}{(a^2 + s^2)} [-a \cos sx + s \sin sx] \right)_0^\infty$$

$$= \left(0 - \frac{1}{(a^2 + s^2)} [(-a)(1) + 0] \right)$$

$$F(s) = \sqrt{\frac{2}{\pi}} \left(\frac{a}{s^2 + a^2} \right)$$

By Parseval's Identity

$$\int_{-\infty}^\infty [F(s)]^2 \, ds = \int_{-\infty}^\infty [f(x)]^2 \, dx$$

$$\frac{2}{\pi} \int_{-\infty}^\infty \left(\frac{a^2}{(s^2 + a^2)^2} \right) \, ds = \int_{-\infty}^\infty \{e^{-a|x|}\}^2 \, dx$$

$$\frac{4}{\pi} \int_0^\infty \left(\frac{a^2}{(s^2 + a^2)^2} \right) \, ds = 2 \int_0^\infty \{e^{-a|x|}\}^2 \, dx$$

$$\frac{4a^2}{\pi} \int_0^\infty \frac{1}{(s^2 + a^2)^2} \, ds = 2 \int_0^\infty e^{-2ax} \, dx = 2 \left(-\frac{e^{-2ax}}{2a} \right)_0^\infty = \frac{1}{a}$$

$$\Rightarrow \int_0^\infty \frac{1}{(s^2 + a^2)^2} \, ds = \frac{\pi}{4a^3}$$

6. Using Parseval's identity of the Fourier cosine transform, evaluate

$$\int_0^{\infty} \frac{\sin ax}{x(a^2 + x^2)} dx \text{ if } f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases} \text{ and } g(x) = e^{-a|x|}, a > 0$$

Solution: The F.C.T of $f(x)$ is $F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$

$$\begin{aligned} &= \sqrt{\frac{2}{\pi}} \int_0^a 1 \cdot \cos sx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{\sin sx}{s} \right)_0^a = \sqrt{\frac{2}{\pi}} \left(\frac{\sin sa}{s} \right) \end{aligned}$$

The F.C.T of $g(x)$ is $G_c(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(x) \cos sx \, dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \cos sx \, dx = \sqrt{\frac{2}{\pi}} \frac{a}{(s^2 + a^2)}$$

By Parseval's Identity,

$$\int_0^{\infty} F_c(s) G_c(s) ds = \int_0^{\infty} f(x) g(x) dx$$

$$\int_0^{\infty} \sqrt{\frac{2}{\pi}} \left(\frac{\sin sa}{s} \right) \sqrt{\frac{2}{\pi}} \frac{a}{(s^2 + a^2)} ds = \int_0^{\infty} e^{-ax} dx = \left(-\frac{e^{-ax}}{a} \right)_0^{\infty}$$

$$\frac{2a}{\pi} \int_0^{\infty} \left(\frac{\sin sa}{s(s^2 + a^2)} \right) ds = \frac{1}{a} (1 - e^{-a^2})$$

$$\int_0^{\infty} \left(\frac{\sin sa}{s(s^2 + a^2)} \right) ds = \frac{\pi}{2a^2} (1 - e^{-a^2})$$

Changing stox, we get

$$\int_0^{\infty} \left(\frac{\sin ax}{x(x^2 + a^2)} \right) dx = \frac{\pi}{2a^2} (1 - e^{-a^2})$$

7. Show that $e^{-\frac{x^2}{2}}$ is self – reciprocal under the Fourier cosine transform.

Solution: The F.C.T of $f(x)$ is

$$F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$$

Here $f(x) = e^{-\frac{x^2}{2}}$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2}} \cos sx \, dx$$

$$= \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2}} e^{isx} dx \{ \because e^{isx} = \cos sx + i \sin sx \}$$

$$= \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2} + isx} dx$$

$$= \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{1}{2}(x^2 - 2isx)} dx$$

$$= \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{1}{2}(x^2 - 2isx + (is)^2)} e^{\frac{s^2}{2}} dx$$

$$= e^{\frac{s^2}{2}} \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{1}{2}(x-is)^2} dx$$

Put $t = x - is \Rightarrow dt = dx$

$$= e^{\frac{s^2}{2}} \text{R.P. of } \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-\frac{(t)^2}{2}} dt$$

$$\text{But } \int_0^{\infty} e^{-\frac{t^2}{2}} dt = \int_0^{\infty} e^{-u} \frac{du}{\sqrt{2}\sqrt{u}}$$

Where $\frac{t^2}{2} = u \Rightarrow t dt = du \Rightarrow dt = \frac{du}{t} = \frac{du}{\sqrt{2}\sqrt{u}}$ Where $t = \sqrt{2}\sqrt{u}$

$$= \frac{1}{\sqrt{2}} \int_0^{\infty} e^{-u} u^{-\frac{1}{2}} du$$

$$= \frac{1}{\sqrt{2}} \Gamma \frac{1}{2}$$

$$= \frac{1}{\sqrt{2}} \sqrt{\pi} = \frac{\sqrt{\pi}}{\sqrt{2}}$$

$$\therefore F_c(s) = R.P. \text{ of } \sqrt{\frac{2}{\pi}} e^{\frac{-s^2}{2}} \frac{\sqrt{\pi}}{\sqrt{2}} = e^{\frac{-s^2}{2}}$$

Since F.C.T of $e^{\frac{-x^2}{2}}$ is $e^{\frac{-s^2}{2}}$, the function $f(x) = e^{\frac{-x^2}{2}}$ is self-reciprocal under the Fourier Cosine transform.

8. Find the Fourier sine transform of $\frac{e^{-ax}}{x}$. Hence find F. S. T of $\frac{1}{x}$.

Solution: The F.S.T of $f(x)$ is $F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin sx \, dx$

Here $f(x) = \frac{e^{-ax}}{x}$

$$I = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} \sin sx \, dx \quad \text{--- (1)}$$

Differentiating (1) w.r.to s , we have

$$\begin{aligned} \frac{dI}{ds} &= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} \frac{\partial}{\partial s} [\sin sx] \, dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} x \cos sx \, dx = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cos sx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left(\frac{e^{-ax}}{a^2 + s^2} (-a \cos sx + s \sin sx) \right)_0^\infty \end{aligned}$$

$$\Rightarrow \frac{dI}{ds} = \sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + s^2} \right)$$

$$\therefore I = \sqrt{\frac{2}{\pi}} \int \left(\frac{a}{a^2 + s^2} \right) ds = a \sqrt{\frac{2}{\pi}} \int \left(\frac{1}{a^2 + s^2} \right) ds$$

$$= a \sqrt{\frac{2}{\pi}} \cdot \frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) + A = \sqrt{\frac{2}{\pi}} \tan^{-1} \left(\frac{s}{a} \right) + A$$

When $s = 0$, $I = 0 \therefore A = 0$

$$\text{Hence } I = \sqrt{\frac{2}{\pi}} \tan^{-1} \left(\frac{s}{a} \right) \quad \text{--- (2)}$$

$$\Rightarrow F_s \left(\frac{e^{-ax}}{x} \right) = \sqrt{\frac{2}{\pi}} \tan^{-1} \left(\frac{s}{a} \right)$$

Put $a = 0$ in (2),

$$F_s \left(\frac{1}{x} \right) = \sqrt{\frac{2}{\pi}} \tan^{-1} \infty = \sqrt{\frac{2}{\pi}} \left(\frac{\pi}{2} \right)$$

$$\therefore F_s \left(\frac{1}{x} \right) = \sqrt{\frac{\pi}{2}}$$

9. Find $f(x)$ if its sine transform is $\frac{e^{-sa}}{s}$. Hence find $F_s^{-1} \left\{ \frac{1}{s} \right\}$.

Solution : By inversion Formula, $F_s^{-1} \{ F_s(s) \} = f(x) = \frac{2}{\pi} \int_0^\infty F_s(s) \sin sx \, ds$

$$\text{Given } F_s(s) = \frac{e^{-sa}}{s}$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^\infty \frac{e^{-sa}}{s} \sin sx \, ds \quad \text{--- (1)}$$

$$\text{Let } I = \frac{2}{\pi} \int_0^{\infty} \frac{e^{-sx}}{s} \sin sx \, ds$$

Differentiating (1), w.r. to x

$$\frac{dI}{dx} = \frac{2}{\pi} \int_0^{\infty} \frac{e^{-sx}}{s} s \cos sx \, ds = \frac{2}{\pi} \int_0^{\infty} e^{-sx} \cos sx \, ds$$

$$= \frac{2}{\pi} \left[\frac{e^{-sx}}{a^2 + x^2} (-a \cos sx + x \sin sx) \right]_0^{\infty}$$

$$\frac{dI}{dx} = \frac{2}{\pi} \left(\frac{a}{a^2 + x^2} \right)$$

$$I = \frac{2}{\pi} \int \frac{a}{a^2 + x^2} dx + A$$

$$f(x) = \frac{2}{\pi} \tan^{-1} \left(\frac{x}{a} \right) + A \quad \left\{ \because \int \frac{a}{a^2 + x^2} dx = \tan^{-1} \left(\frac{x}{a} \right) \right\}$$

$$\text{Put } x = 0, f(0) = 0 + A = A$$

$$\text{But from (1), } f(0) = 0$$

$$\therefore A = 0$$

$$\text{Hence } f(x) = F_s^{-1} \left\{ \frac{e^{-sa}}{s} \right\} = \frac{2}{\pi} \tan^{-1} \left(\frac{x}{a} \right)$$

Put $a = 0$ in the above result,

$$f(x) = F_s^{-1} \left\{ \frac{1}{s} \right\} = \frac{2}{\pi} \tan^{-1} \infty = \frac{2}{\pi} \left(\frac{\pi}{2} \right) = 1$$

$$\therefore F_s^{-1} \left\{ \frac{1}{s} \right\} = 1$$

10. Find the Fourier cosine transform of e^{-ax} and hence deduce that

$$\int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-ma}$$

$$\text{Solution : The F.C.T of } f(x) \text{ is } F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx \, dx$$

$$\text{Here } f(x) = e^{-ax}$$

$$\begin{aligned} F_c(s) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \cos sx \, dx \\ &= \sqrt{\frac{2}{\pi}} \frac{a}{s^2 + a^2} \end{aligned}$$

$$\text{By inverse F.C.T } f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c(s) \cos sx \, ds$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \frac{a}{s^2 + a^2} \cos sx \, ds$$

$$e^{-ax} = \frac{2}{\pi} \int_0^{\infty} \frac{a}{s^2 + a^2} \cos sx \, ds \quad (\because f(x) = e^{-ax})$$

$$\frac{\pi}{2a} e^{-ax} = \int_0^{\infty} \frac{1}{s^2 + a^2} \cos sx \, ds$$

$$\Rightarrow \int_0^{\infty} \frac{\cos sx}{s^2 + a^2} ds = \frac{\pi}{2a} e^{-ax}$$

Changing s to x and x to m We get

$$\int_0^{\infty} \frac{\cos mx}{x^2 + a^2} ds = \frac{\pi}{2a} e^{-ma}$$

11. Find the Fourier sine transform of $f(x) = \frac{1}{x}$.

Solution:

The F.S.T of $f(x)$ is

$$F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx \, dx = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{1}{x} \sin sx \, dx$$

$$\text{Let } sx = \theta \Rightarrow s dx = d\theta$$

$$\begin{aligned}
&= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{s}{\theta} \sin \theta \frac{d\theta}{s} \\
&= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\sin \theta}{\theta} d\theta = \sqrt{\frac{2}{\pi}} \left(\frac{\pi}{2} \right) \left(\because \int_0^\infty \frac{\sin \theta}{\theta} d\theta = \frac{\pi}{2} \right) \\
&= \sqrt{\frac{\pi}{2}}
\end{aligned}$$

12. Find the Fourier cosine transform of e^{-x^2}

Solution: The F.C.T of $f(x)$ is

$$F_c(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos sx \, dx$$

Here $f(x) = e^{-x^2}$

$$\text{Let } I = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos sx \, dx$$

$$I = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x^2} \cos sx \, dx$$

$$= R.P \int_0^\infty e^{-x^2} e^{isx} \, dx$$

$$= R.P \int_0^\infty e^{-x^2 + isx} \, dx$$

$$= R.P \int_0^\infty e^{-(x^2 - isx)} \, dx \quad \text{--- (1)}$$

Since $a^2 - 2ab + b^2 = (a - b)^2$

$$(x^2 - isx) = -\frac{1}{2} \left[(x)^2 - 2 \frac{is}{2} x + \left(\frac{is}{2} \right)^2 - \left(\frac{is}{2} \right)^2 \right]$$

$$= - \left[\left(x - \frac{is}{2} \right)^2 + \frac{s^2}{4} \right] = - \left(x - \frac{is}{2} \right)^2 + \frac{s^2}{4}$$

$$\text{Now (1) becomes } I = R.P \int_0^\infty e^{-\left(x - \frac{is}{2} \right)^2 + \frac{s^2}{4}} \, dx$$

$$= R.P \int_0^\infty e^{-\left(x - \frac{is}{2} \right)^2} e^{\frac{s^2}{4}} \, dx$$

$$= R.P \sqrt{\frac{2}{\pi}} e^{\frac{s^2}{4}} \int_0^\infty e^{-\left(x - \frac{is}{2} \right)^2} \, dx$$

$$\text{Put } x - \frac{is}{2} = t \Rightarrow dx = dt$$

$$I = R.P \sqrt{\frac{2}{\pi}} e^{\frac{s^2}{4}} \int_0^\infty e^{-t^2} \, dt \quad \text{--- (2)}$$

$$\text{Put } t^2 = u \Rightarrow 2t \, dt = du \text{ and } t = \sqrt{u} \Rightarrow dt = \frac{du}{2t} = \frac{du}{2\sqrt{u}}$$

$$I = R.P \sqrt{\frac{2}{\pi}} e^{\frac{s^2}{4}} \int_0^\infty e^{-u} \frac{du}{2\sqrt{u}}$$

$$= R.P \sqrt{\frac{2}{\pi}} e^{\frac{s^2}{4}} \int_0^\infty e^{-u} \frac{du}{2\sqrt{u}}$$

$$= R.P \sqrt{\frac{2}{\pi}} \frac{e^{\frac{s^2}{4}}}{2} \int_0^\infty e^{-u} u^{-\frac{1}{2}} \, du$$

$$= R.P \sqrt{\frac{2}{\pi}} \frac{e^{\frac{s^2}{4}}}{\sqrt{2}\sqrt{2}} \int_0^\infty e^{-u} u^{\frac{1}{2}-1} \, du$$

$$= R.P \sqrt{\frac{1}{2\pi}} e^{\frac{s^2}{4}} \Gamma\left(\frac{1}{2}\right)$$

$$[\text{We know that } \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \Rightarrow \Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} e^{-u} u^{-\frac{1}{2}} du]$$

$$= R.P \sqrt{\frac{1}{2\pi}} e^{\frac{s^2}{4}} \sqrt{\pi} \quad (\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi})$$

$$= \frac{1}{\sqrt{2}} e^{\frac{s^2}{4}}$$

Therefore F. C. T of e^{-x^2} is $\frac{1}{\sqrt{2}} e^{-\frac{s^2}{4}}$

Problems for Practice

1. Using Fourier cosine transform method, evaluate $\int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)}$

$$\text{Ans : } \frac{\pi}{2ab(a+b)}$$

2. Find the Fourier sine transform of $f(x) = \begin{cases} x^2, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$

$$\text{Ans : } \sqrt{\frac{2}{\pi}} \left(\frac{(2 - s^2)\cos s + 2s\sin s - 2}{s^3} \right)$$

UNIVERSITY QUESTIONS

PART - A

MAY 2017

1. If $F(s)$ is the Fourier transform of $f(x)$, Prove that $F[f(x - a)] = e^{isa}F(s)$
2. Find the Fourier sine transform of $\frac{1}{x}$.

PART - B

1. Find the Fourier transform of $f(x) = \begin{cases} 1 & |x| < 2 \\ 0 & |x| > 2 \end{cases}$. Hence find $\int_0^\infty \frac{\sin x}{x} dx$. And $\int_0^\infty \frac{\sin^2 x}{x^2} dx$.
2. Find the Fourier transform of $e^{-a^2 x^2}$, $a > 0$
3. Evaluate $\int_0^\infty \frac{ds}{(s^2+1)(s^2+4)}$ using Fourier transform.

PART - A

MAY 2016

1. State Fourier Integral theorem.
2. Find the Fourier transform of a derivative of the function $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$.

PART - B

1. Find the Fourier sine and cosine transform of $f(x) = e^{-ax}$ for $x \geq 0$ and $a > 0$. Hence deduce the integrals $\int_0^\infty \frac{\cos sx}{a^2 + s^2} ds$ and $\int_0^\infty \frac{s \sin sx}{a^2 + s^2} ds$.
2. (a) Find the Fourier transform of $f(x) = e^{-\frac{x^2}{2}}$ in $(-\infty, \infty)$.
3. (b) Find the Fourier transform of $f(x) = 1 - |x|$ if $|x| < 1$ and hence find the value of $\int_0^\infty \frac{\sin^4 t}{t^4} dt$.

PART - A

DEC 2016

1. State change of scale property on Fourier transforms.
2. Find the infinite Fourier sine transform of $f(x) = \frac{1}{x}$.

PART - B

1. Find the Fourier Transform of $f(x) = \begin{cases} 1 - |x| & |x| < 1 \\ 0 & |x| > 1 \end{cases}$ and hence deduce that $\int_0^\infty \left[\frac{\sin t}{t} \right]^4 dt = \frac{\pi}{3}$.
2. Find the infinite Fourier transform of $f(x) = \frac{e^{-ax}}{x}$ hence deduce the infinite Fourier sine transform of $\frac{1}{x}$.
3. (i) Find the Fourier transform of $e^{-a^2 x^2}$, $a > 0$. Hence Show that $e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier transform.

PART-A

DEC 2015

1. If $F(s)$ is the Fourier transform of $f(x)$, Prove that $F\{f(ax)\} = \frac{1}{|a|} F\left(\frac{s}{a}\right)$ where $a \neq 0$
2. Evaluate $\int_0^\infty \frac{s^2 ds}{(s^2 + a^2)(s^2 + b^2)}$ using Fourier transform.

PART-B

1. (i) Find the Fourier sine integral representation of the function $f(x) = e^{-x} \sin x$.
(ii) Find the Fourier cosine transform of the function $\frac{e^{-ax} - e^{-bx}}{x}$, $x > 0$
2. (i) Find the Fourier transform of $f(x) = \begin{cases} 1 - |x| & |x| \leq 1 \\ 0 & |x| > 1 \end{cases}$. Hence evaluate i) $\int_0^\infty \frac{\sin^4 t}{t^4} dt$

(ii) Verify the Convolution theorem for Fourier transform $f(x) =$

$$g(x) = e^{-x^2}$$

PART-A

MAY 2015

1. If $F\{f(x)\} = F(s)$, then $F\{f(x-a)\} = e^{isa}F(s)$
2. Find the Fourier Sine transform of $\frac{1}{x}$

PART-B

1. Find the Fourier transform of $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$. Hence find

$$\int_0^\infty \frac{\sin x}{x} dx. \text{ Using Parseval's identity Prove that } \int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

2. (i) Find the Fourier transform of $e^{-a^2 x^2}$, $a > 0$. Hence Show that $e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier transform.
(ii) Find the Fourier Cosine transform of x^{n-1}

PART-A

DEC 2014

1. Prove that $F[f(x-a)] = e^{isa}F(s)$
2. Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$

PART-B

1. Show that the transform of $e^{-x^2/2}$ is $e^{-s^2/2}$ by finding the Fourier transform of $e^{-a^2 x^2}$, $a > 0$
2. Find the F.S.T and F.C.T of e^{-ax} , $a > 0$ and hence deduce the inversion Formula

3. Find the Fourier transform of $e^{-a|x|}$, $a > 0$ and hence deduce that

$$F\{xe^{-a|x|}\} = i\sqrt{\frac{2}{\pi}} \frac{2as}{(s^2+a^2)^2}$$

4. Solve for $f(x)$ from the integral equation $\int_0^\infty f(x) \cos ax \, dx = e^{-a}$

PART-A

MAY 2014

1. Find the Fourier sine transform of $f(x) = \frac{1}{x}$
2. State the Fourier integral Theorem

PART-B

1. Find the Fourier transform of $e^{-a|x|}$, $a > 0$ and hence deduce that

$$(i) \int_0^\infty \frac{\cos xt}{\alpha^2+t^2} dt = \frac{\pi}{2\alpha} e^{-a|x|} \quad (ii) F\{xe^{-a|x|}\} = i\sqrt{\frac{2}{\pi}} \frac{2as}{(s^2+a^2)^2},$$

2. Solve $f(x)$ from the integral equation $\int_0^\infty f(x) \sin sx \, dx = \begin{cases} 1, & 0 \leq s < 1 \\ 2, & 1 \leq s < 2 \\ 0, & s \geq 2 \end{cases}$
3. Find the Fourier transform of $f(x) = \frac{1}{\sqrt{|x|}}$
4. Using Parseval's identity, Evaluate i) $\int_0^\infty \frac{dx}{(a^2+x^2)^2}$ ii) $\int_0^\infty \frac{x^2 dx}{(a^2+x^2)^2}$ where $a > 0$

PART-A

DEC 2013

1. Define self reciprocal with respect to Fourier Transform
2. Prove that $F[f(x-a)] = e^{ias}F[f(x)]$

PART-B

1. Find the Fourier transform of $f(x) = \begin{cases} 1-x^2|x| < 1 \\ 0, & |x| > 1 \end{cases}$. Hence show that
i) $\int_0^\infty \frac{\sin s - s \cos s}{s^3} ds = \frac{3\pi}{16}$ ii) $\int_0^\infty \frac{(x \cos x - \sin x)^2}{x^6} dx = \frac{\pi}{15}$
2. Using Fourier cosine Transform evaluate $\int_0^\infty \frac{dx}{(x^2+a^2)^2}$

3. Find the function whose Fourier sine transform is $\frac{e^{-as}}{s}$, $a > 0$

PART-A

MAY-2013

1. Find the Fourier sine Transform of e^{ax}
2. If $F[f(x)] = F(s)$, Prove that $F[f(ax)] = \frac{1}{a}F\left(\frac{s}{a}\right)$

PART-B

1. Find the Fourier transform of $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$. Hence find

$$\int_0^\infty \frac{\sin x}{x} ds$$

2. Verify convolution theorem under Fourier Transform for

$$f(x) = g(x) = e^{-x^2}$$

3. Obtain the Fourier Transform of $e^{-x^2/2}$.